

EXPERIMENT: 5

A python program to implement multi layer perceptron with back propagation

Aim:

To implement a Multilayer Perceptron (MLP) model using a given dataset and to classify the output based on input features using a neural network.

Procedure:

1. Import required libraries and load the dataset.
2. Split data into features (X) and target (y), then apply train-test split.
3. Normalize data using feature scaling.

4. Forward Propagation:

Compute outputs using:

- $Z = WX + b$
- $A = f(Z)$ (activation function like ReLU/Sigmoid)

5. Backpropagation:

- Error = $y - \hat{y}$
- Update weights:

$$W = W + \eta \cdot \nabla W$$

6. Train the MLP model using training data.
7. Predict output for test data.
8. Evaluate using accuracy and confusion matrix.

CODE:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
```

```

from sklearn.preprocessing import StandardScaler

from sklearn.neural_network import MLPClassifier

from sklearn.metrics import accuracy_score, confusion_matrix

data = pd.read_csv("MLP.csv")

print("Dataset Preview:")

print(data.head())

X = data.iloc[:, :-1]

y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)


mlp = MLPClassifier(hidden_layer_sizes=(6,6),

                    activation='relu',

                    learning_rate_init=0.01,

                    max_iter=500)

mlp.fit(X_train, y_train)

y_pred = mlp.predict(X_test)

print("\nAccuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

plt.figure()

plt.scatter(X_test[:,0], X_test[:,1], c=y_pred)

plt.xlabel("Feature 1")

plt.ylabel("Feature 2")

```

```
plt.title("MLP Predictions")

plt.show()

plt.figure()

plt.plot(mlp.loss_curve_)

plt.xlabel("Iterations")

plt.ylabel("Loss")

plt.title("Loss Curve")

plt.show()
```

OUTPUT:

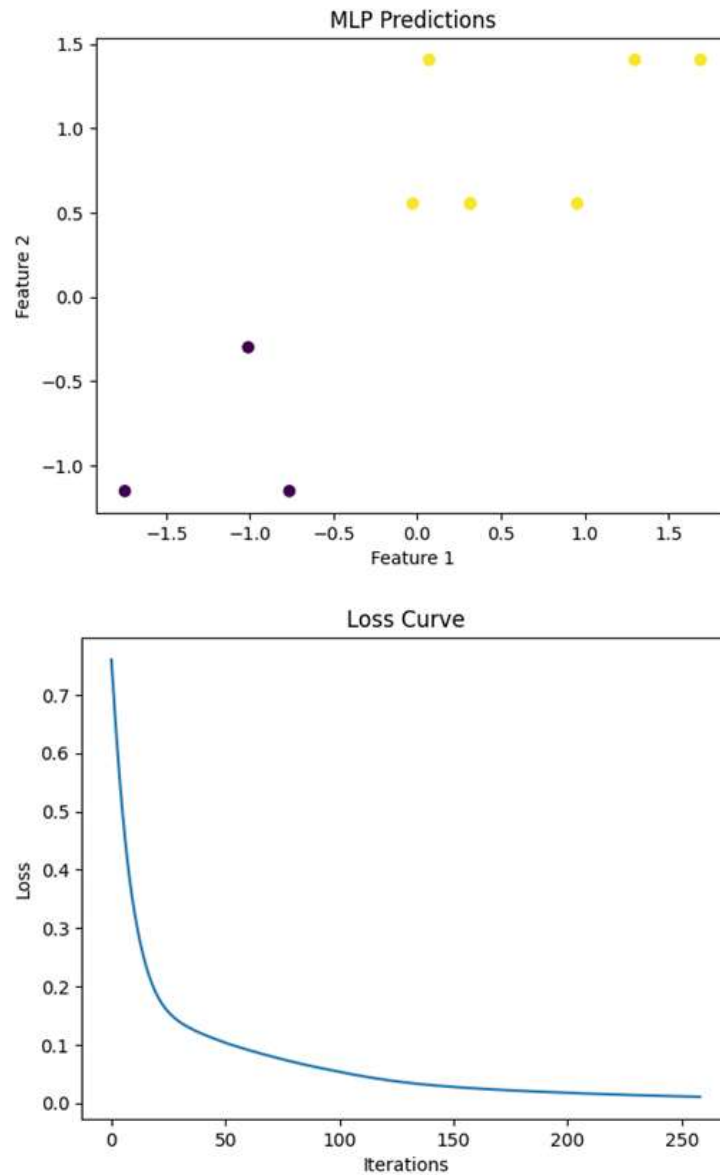
Dataset Preview:

	StudyHours	SleepHours	PracticeTests	ExamResult
0	1.0	5	0	0
1	2.0	6	1	0
2	1.5	5	0	0
3	3.0	6	1	0
4	2.5	6	1	0

Accuracy: 1.0

Confusion Matrix:

```
[[3 0]
 [0 7]]
```



RESULT:

The Multilayer Perceptron (MLP) model was successfully trained and achieved good accuracy in classifying the data.